

UZ2400

U-Power1000 Module User Manual

Application Note

AN-2400-14

The content of this technical document is subject to change without notice. Please contact UBEC for further information.

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1. General Information

The U-Power1000 module is an IEEE 802.15.4 ZigBee™ compliant solution that satisfies the requirements of low-power and long-range wireless sensor applications. It integrates UBEC's UZ2400 2.4GHz transceiver with UBEC's amplifiers to extend the transmitting and the receiving distance at low current consumption. Both the low current consumption and long distance capability are especially suitable for battery-limited applications. The module's flexible and reliable interface allows the users to develop configurations appropriate for their applications.

The U-Power1000 module, derived from UBEC's U-Force module, operates within the ISM 2.4 – 2.5 GHz frequency band and contains UBEC's UZ2400, UP2206, UA2725 and other necessary components such as crystal oscillator, switches, inductors and capacitors. The 802.15.4 compliant transceiver within the UZ2400 chip features a maximum of 5 MHz serial interface SPI bus for control and data transfer. It is a simple SPI interface slave device consisting of a 4-wire bus (SCLK, SI, SO, and SEN) that provide accesses to various subunits such as MAC/BB/RF control/status registers, TXFIFOs, RXFIFOs and security key table.

The UP2206 is a high-efficiency, high-output, linear power amplifier (PA). It provides the necessary power output to meet the long distance requirements demanded by certain applications. The UA2725 is a high performance, low noise amplifier (LNA) with a noise figure as low as 1.8dB at 2.4GHz. The use of an LNA in a link can also extend the communication distance.

2. Features

- ☐ 2.4GHz IEEE 802.15.4 and ZigBee Compliant
- ☐ 3.0~3.6V Operation
- ☐ >1000m Range (Environment dependent)
- ☐ SMA Connector
- ☐ 2 GSG(ground-signal-ground) interfaces provided
- ☐ Receiver Sensitivity: -101dBm
- ☐ Tx Power : 22dBm
- ☐ Tx Current : 154mA
- ☐ Rx Current : 32mA
- ☐ Size: 36mmx14mm
- ☐ Shielding Case Option

* ZigBee™ is a trade mark of ZigBee Alliance.

3. PCB of U-Power1000 Module

Two types of the U-Power1000 module are available, depending on the antenna configurations.

Figure 1 shows the top view of a U-Power1000 module with a SMA connector. The PCB size is 36mm x 14 mm.



Figure 1. UZ2400 U-Power1000 Module

4. U-Power1000 Pin Configuration

UZ2400 U-Power1000 module pin assignments and their functions are described below.

Pin Number	Pin Name	Type	Description
1	GPIO0	DIO	General purpose digital I/O, also used as an external PA enable
2	GPIO1	DIO	General purpose digital I/O, also used as an external TX/RX switch control
3	GPIO5	DIO	General purpose digital I/O
4	GPIO4	DIO	General purpose digital I/O
5	GPIO2	DIO	General purpose digital I/O, also used as an external TX/RX switch control
6	GPIO3	DIO	General purpose digital I/O
7	RESETN	DI	Global hardware reset pin, active low
8	WAKE	DI	External wake up trigger, active high / low can be programmable.
9	INT	DO	Interrupt pin to micro-processor : Level trigger, Hi / Low programmable
10	SO	DIO	Serial interface data output from UZ2400 or I2C clock
11	SI	DIO	Serial interface data input to UZ2400 or I2C data in/out
12	SCLK	DI	Serial interface clock
13	SEN	DI	Serial interface enable
14	CLKOUT	DIO	20/10/5/2.5 MHz Clock output
15	XTAL32N	AI	32 kHz Crystal input (-) for internal RTC used
16	XTAL32P	AI	32 kHz Crystal input (+) for internal RTC used
17	GND	Ground	Ground

18	VCC	Power	3V
19	VCC	Power	3V
20	GND	Ground	Ground

UZ2400 U-Power1000 module pin names and pin number allocation map are shown in Figure 2.

It can interface with other devices such as a sensor, an LED, a host controller, a push button, a joystick or power relays through GPIO signals.

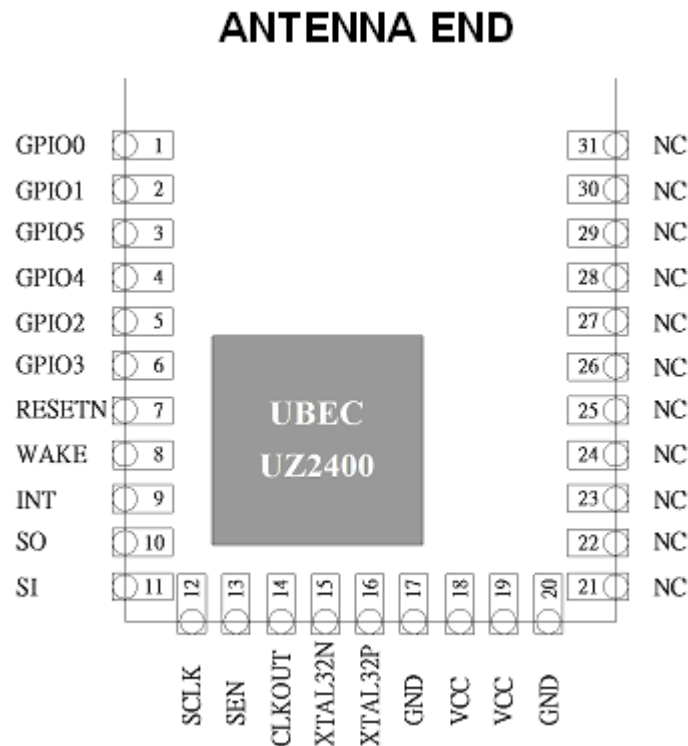


Figure 2. U-Power1000 Module Pin Name Location Map

5. Electrical Specifications

ITEM	SPECIFICATION	UNIT
Frequency range	ISM band 2.4 – 2.5	GHz
Input voltage	3	V
Current consumption	TX :154 RX : 32	mA
TX Output power	19(Typ)/22(Max.)	dBm
TX EVM	10% ~15% (2405 ~ 2480 MHz)	%
RX sensitivity	-101	dBm

For detailed functional descriptions and electrical characteristics of the UZ2400 chip, please refer to UZ2400 datasheet.

5.1. U-Power1000 RF module TX output power

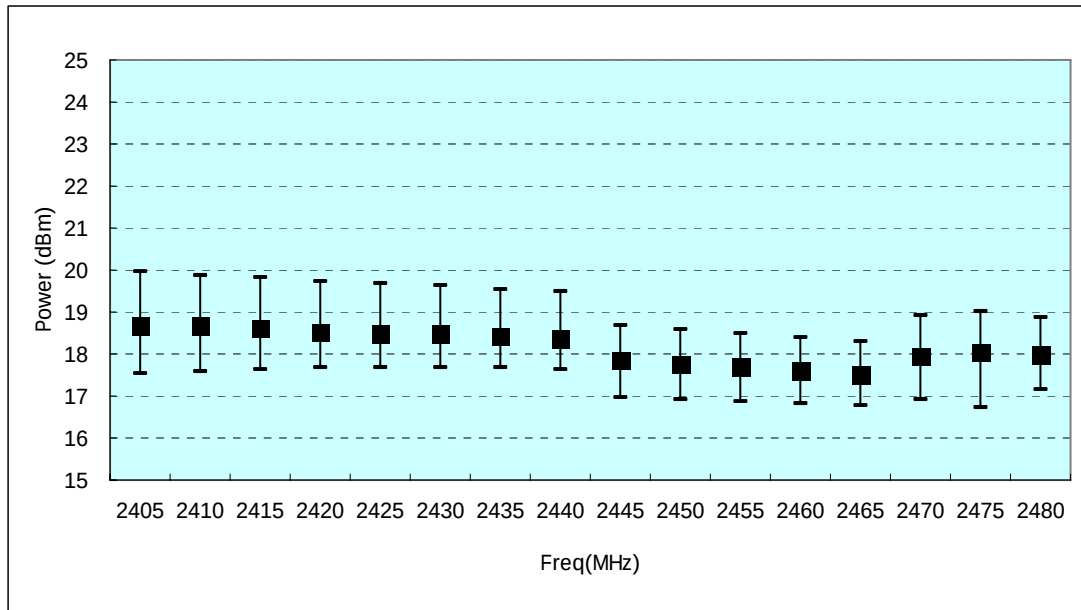


Figure 3. Typical U-Power1000 RF Module TX Output Power Test Results with Pin = -10dBm

5.2. TX EVM

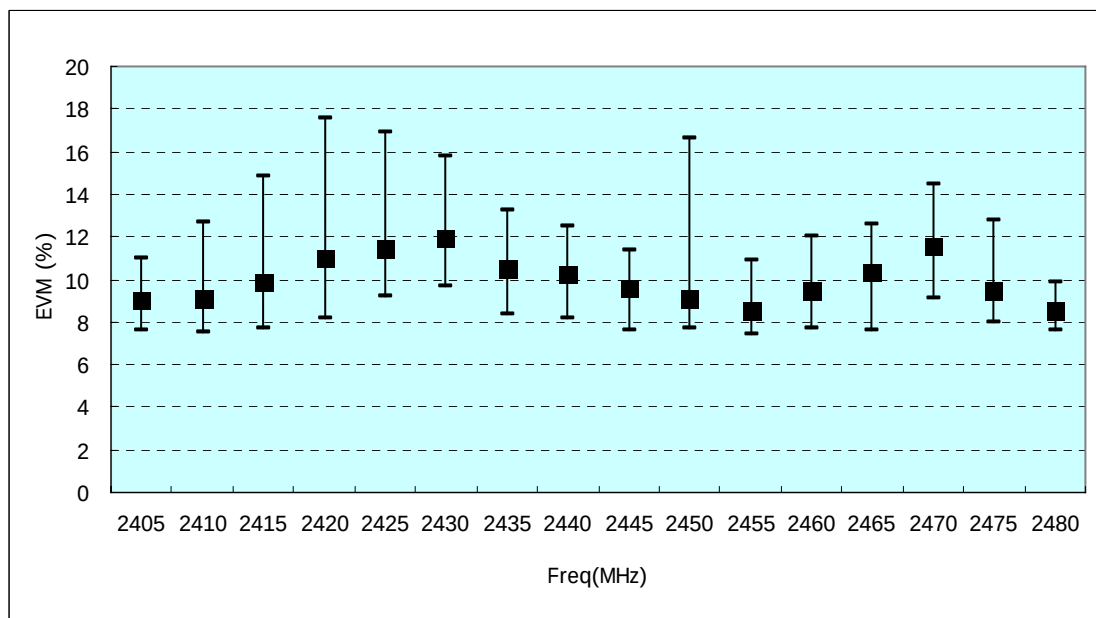


Figure 4. Typical U-Power1000 RF Module TX EVM Test Results

5.3. RX sensitivity

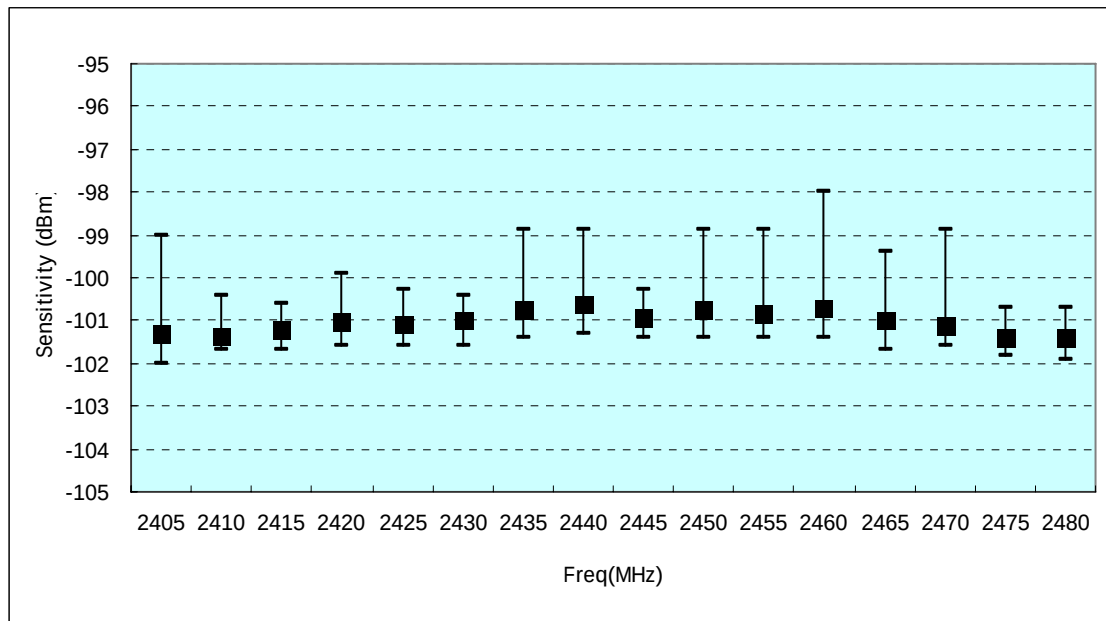


Figure 5. Typical U-Power1000 RF Module RX Sensitivity Test Results

5.4. LOS (Line-of-Sight) Result

Software: UBEC developed program:19TM06

Platform: Window XP/ASUS Notebook

Module: U-Power1000

Criteria: PER (Packet-Error-Rate)<5%

- ☐ Google Real Estate Search
- ☐ Photos from Picasa Web Albums
- ☐ Photos from Panoramio
- ☐ Gas Prices from GasBuddy.com
- ☒ Distance Measurement Tool
- ☐ Browse Popular Maps

Distance Measurement Tool

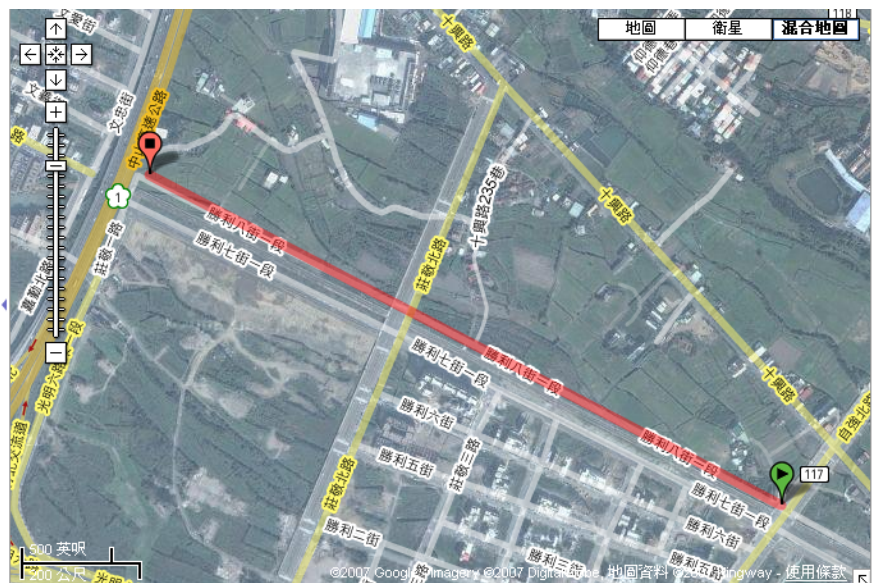
在地圖上按一下，以追蹤您想要測量的路徑。

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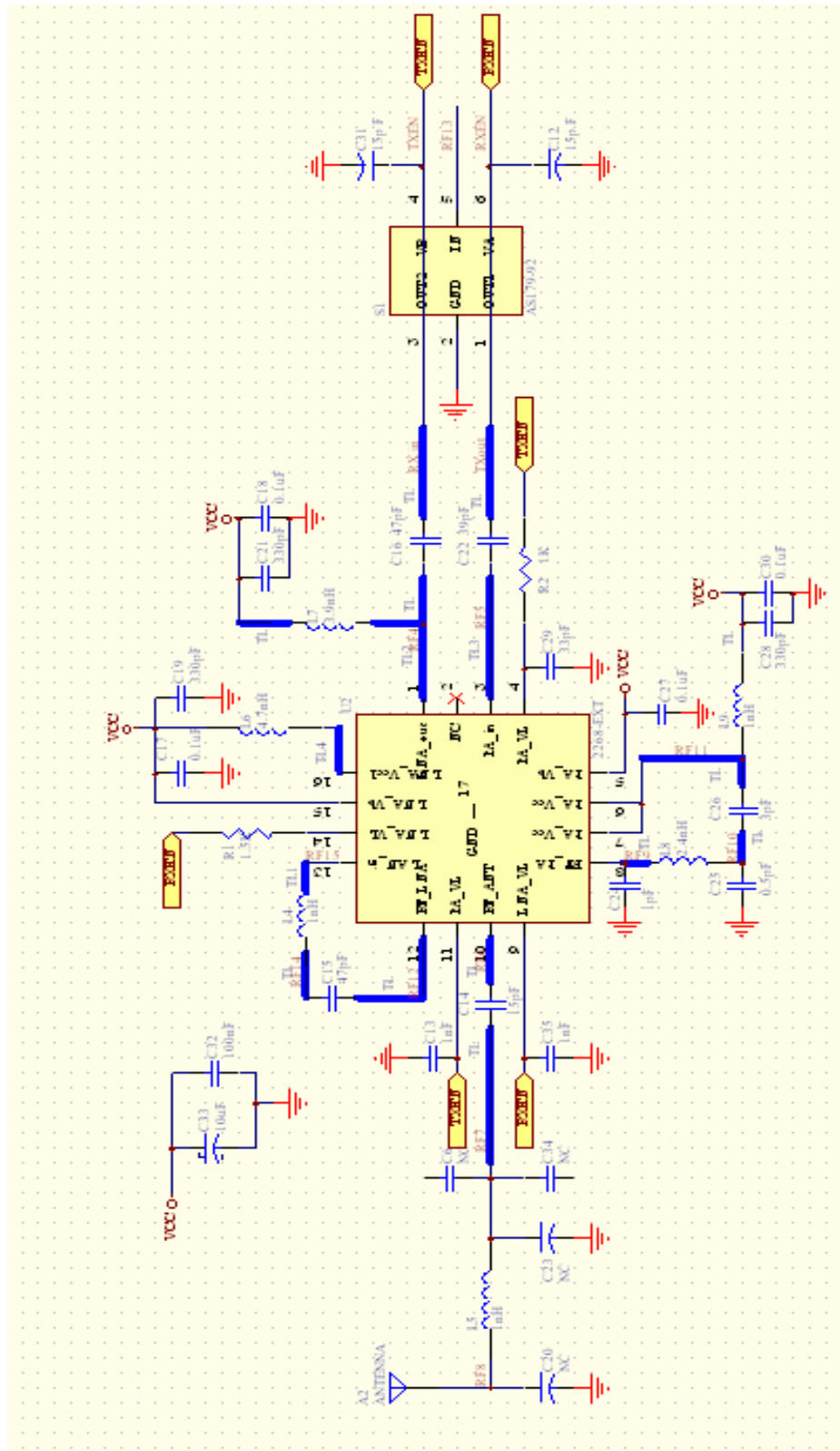
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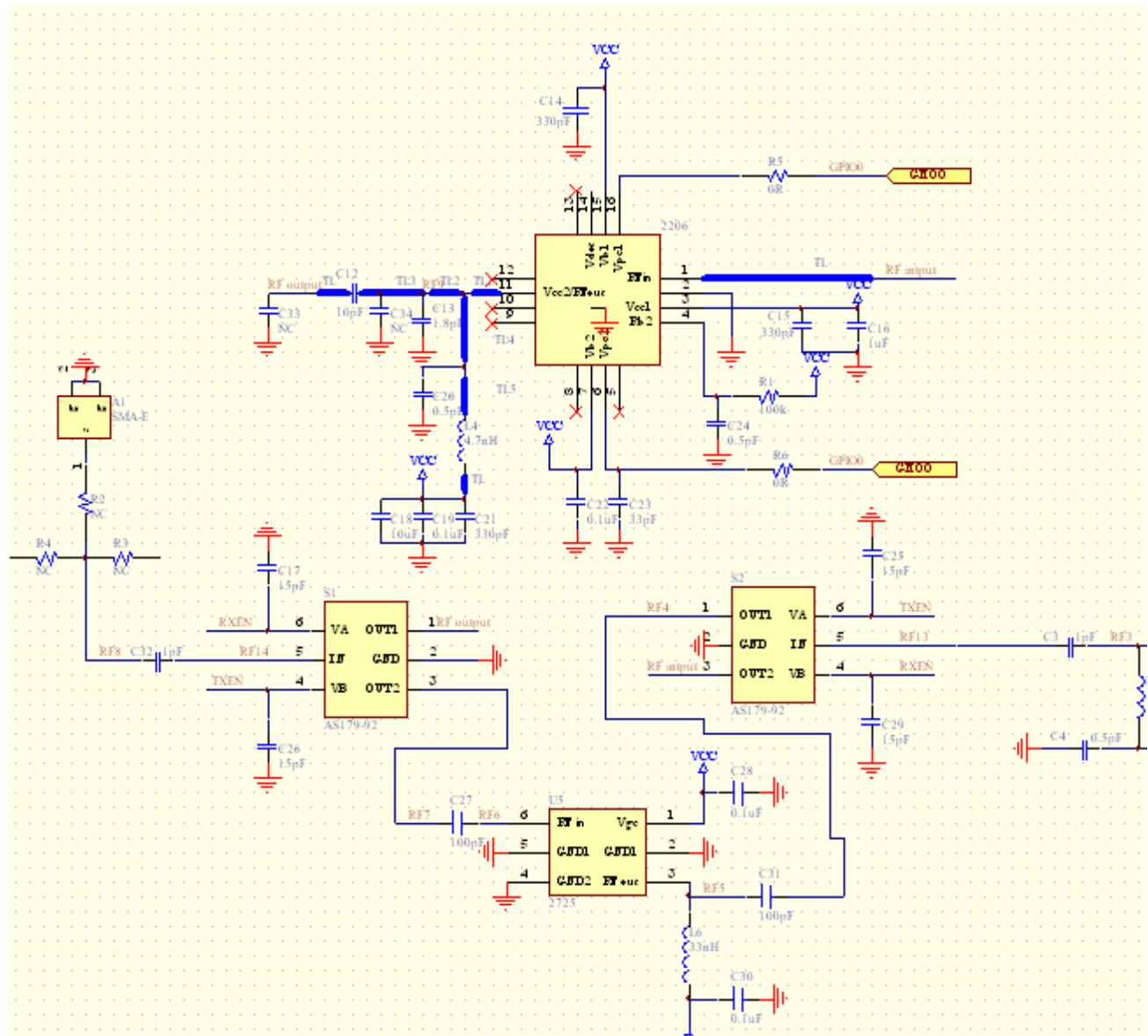
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6. Schematic





7. Mechanical Dimension Drawing

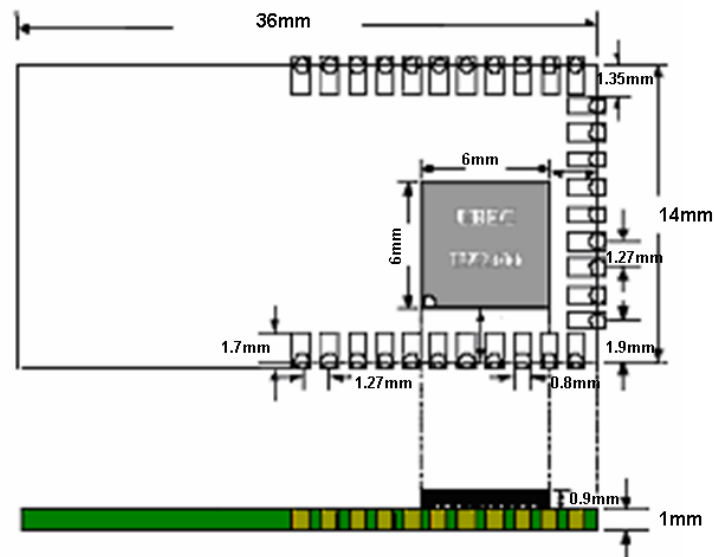
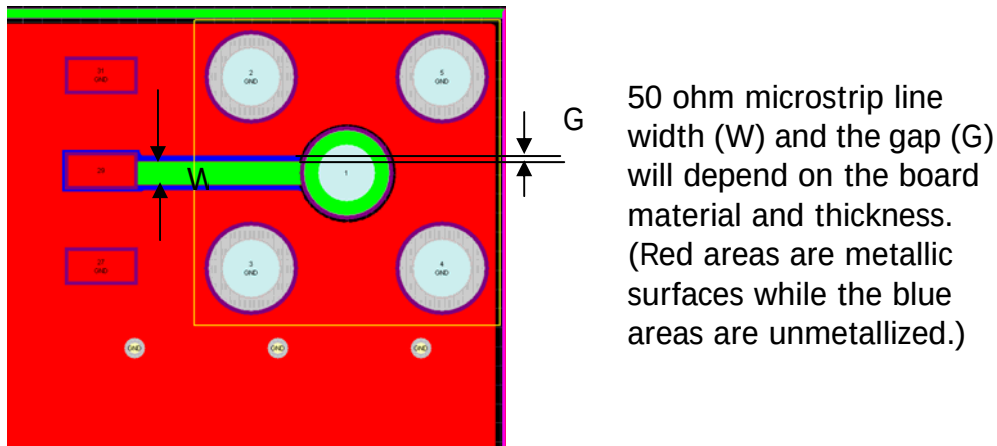


Figure 6. U-Power1000 Module Dimensions

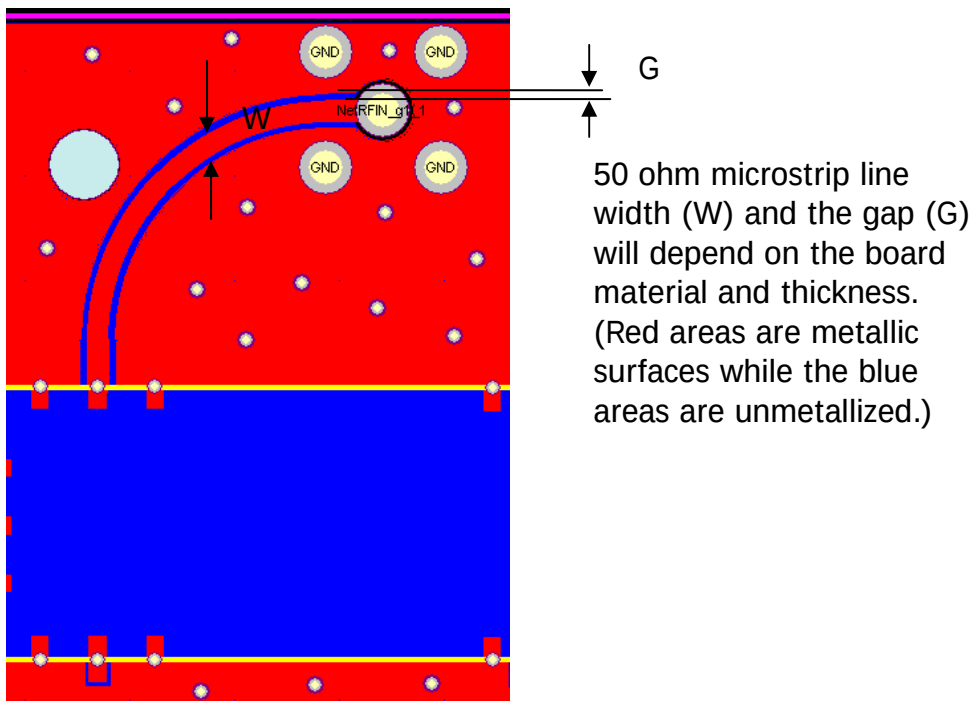
8. Ground-Signal-Ground (GSG) Interface Design

To prevent a huge RF power loss from the GSG to the system board, the recommended rules for laying the micro-strip lines should be obeyed. Please refer to UBEC's Application Note: AN-PCB1-01.

8.1. For 4-layer PCB



8.2. For 2-layer PCB



Revision History

Revision	Date	Description of Change
0.0	2007/10/30	Initial version

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